

DEAKIN UNIVERSITY

INTRODUCTION TO DATA SCIENCE AND ARTIFICIAL INTELLIGENCE

ONTRACK SUBMISSION

Task P2

Submitted By:

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SIT112 - TASK COMPLETION REPORT

Important note 1: Please **do not** include Python code in this report. It would be acceptable, though, to make brief references to function names or different parts of the task's notebook when needed.

Important note 2: Please **do not** leave any response field empty; if not applicable, enter NA.

1. TASK SUMMARY

TASK NAME (ABBREVIATION)	YOUR NAME (STUDENT ID)	YOUR DEAKIN EMAIL
Pass Task P2	Jahan Garg (226100103)	s226100103@deakin.edu.au

2. TASK DESCRIPTION

ITEM	RESPONSE
What was the objective(s) of the task?	The objective of this task was to apply core pandas concepts, covered in the Week 2 lecture on Essentials for Data Analysis, to a real-world dataset, in order to read, examine, transform, and analyse structured tabular data, and to draw meaningful conclusions from grouped and aggregated results.
What kind of data did you work with?	The task involved working with a single CSV file, avocado.csv, sourced from the Kaggle Avocado Prices dataset, containing 18,249 records of historical avocado prices and sales volumes across 54 U.S. regions between 2015 and 2018. The dataset included fields such as Date, AveragePrice, Total Volume, PLU-coded sales volumes (4046, 4225, 4770), bag size categories (Total, Small, Large, and XLarge Bags), avocado type (conventional or organic), year, and region.
Briefly describe the data science task you worked on.	The data science task followed a structured pipeline: reading the CSV file into a DataFrame using read_csv(), examining its structure and quality with info() and nunique(), and accessing subsets of the data through bracket notation, dot notation, and the loc[] accessor, consistent with the techniques demonstrated in the Week 2 lecture notebook. A new column, Estimated Revenue, was derived by multiplying Total Volume and AveragePrice. The data was then grouped by region and type to compute average prices, reshaped using pivot() to compare prices across regions in wide format, and grouped by type and year to calculate the standard deviation of Total Bags sold, in order to assess the consistency of sales volumes over time. Bar plots were used to visualise summary statistics of Total Volume by year.

3. TECHNICAL SKILLS

ITEM	RESPONSE
What technical skills did you use during the task?	During this task, I applied a range of pandas techniques, including reading data from a CSV file using <code>read_csv()</code> , inspecting data structure and quality with <code>info()</code> and <code>nunique()</code> , and accessing data through bracket notation, dot notation, and the <code>loc[]</code> accessor. I also performed column arithmetic to create a new Estimated Revenue column, used <code>groupby()</code> combined with <code>mean()</code> and <code>std()</code> to summarise data by region, type, and year, reshaped grouped data using <code>pivot()</code> for regional comparison, and created bar plots using <code>plot.bar()</code> to visualise summary statistics.
List any challenges or obstacles you faced while working on the task and how you overcame them.	I did not encounter major technical obstacles while completing this task, as I had prior exposure to Jupyter Notebook and the pandas library through my Data Capture Technologies unit. Where I was uncertain about exact syntax, I referred to Week 2 unit site material and relevant external sources to confirm correct usage. All cells executed without errors, and I cross-verified my results against the provided P2-Solution notebook after completing my own attempt, to check that my logic and outputs aligned with the expected approach

4. DATA CLEANING AND PREPARATION (ENTER NA WHEN NOT APPLICABLE)

ITEM	RESPONSE
What steps did you take to clean and prepare the data?	As the Unnamed: 0 column and the three PLU code columns (4046, 4225, 4770) were identified in the task instructions as irrelevant to this analysis, I did not remove them from the DataFrame but simply excluded them from all subsequent selections, groupings, and calculations. Data cleaning in this task therefore focused on verifying data quality using <code>info()</code> and <code>nunique()</code> , which confirmed that all 18,249 records were complete with no missing values across any of the 14 columns.
Did you encounter any issues with the data during this process? How did you address these issues with the data?	No missing values or data type issues were identified during this process, as confirmed by <code>info()</code> , which reported 18,249 non-null entries for every column. However, care was needed when interpreting the region column, since some regions represented aggregated totals (for example, Total US) that overlap with the other individual regional entries. As the task instructions specified that overlapping regions should not be accounted for separately, this was noted but not adjusted within the scope of this task.

5. DATA ANALYSIS (ENTER NA WHEN NOT APPLICABLE)

ITEM	RESPONSE
How did you analyse the data?	The data was analysed primarily through grouping and aggregation. I used <code>groupby()</code> on region and type to calculate average prices, then applied <code>pivot()</code> to reshape this grouped data into a wide format for direct comparison across regions. Separately, I grouped the data by type and year to calculate the standard deviation of Total Bags sold, in order to assess how sales volume consistency changed over the four years.
Did you use any visualisation techniques to better understand the data?	Yes, I created a bar plot using <code>plot.bar()</code> to visualise the mean, median, and standard deviation of Total Volume grouped by year, which helped illustrate how sales volume and its variability changed from 2015 to 2018.
What insights did you gain from this analysis?	This analysis revealed several notable patterns. Organic avocados were consistently priced higher than conventional ones, averaging 1.65 compared to 1.16, reflecting a typical premium seen in organic produce markets. Regional variation was also evident, with Houston recording the lowest average organic price (1.27) among all 54 regions, lower than I initially expected given that organic produce is generally more uniformly priced across markets. Finally, the standard deviation of Total Bags sold increased every year from 2015 to 2018 for both avocado types, rising from roughly 7.5×10^5 to 2.0×10^6 for conventional and from 2.6×10^4 to 1.4×10^5 for organic. This indicates that avocado sales volumes became progressively less consistent over time rather than stabilising as the market matured.

6. BASIC REQUIREMENTS FOR THE TASK

ITEM	RESPONSE (YES/NO)
Are you confident in executing the Python code in this task and explaining the output?	Yes
Are you confident in explaining what each line of code does and how it contributes to the solution(s)?	Yes
Are you confident in rewriting or modifying the code after completing this task? - For pass tasks: with guidance, no time limit. - For credit tasks: with limited guidance, no time limit. - For distinction tasks: independently, no time limit. - For high distinction tasks: independently, in a limited time.	Yes

7. CODE ATTACHMENT (NOT APPLICABLE TO THE PASS TASKS – ENTER NA)

ITEM	RESPONSE (YES/NO/NA)
Have you attached the notebook file that contains your solutions (Python code) for this task?	NA
Have you executed all the cells in your attached notebook and ensured there is no error? <i>Please note your submission will not be flagged as complete if your attached notebook contains any error.</i>	NA

8. VIDEO ATTACHMENT (NOT APPLICABLE TO THE PASS/CREDIT TASKS: NA)

ITEM	RESPONSE (THE VIDEO LINK/NA)
Provide the link to the video recording that presents your completed task. This is only for Distinction and High Distinction tasks. Enter NA for Pass/Credit tasks.	NA

9. AI REFLECTION STATEMENT

How did you use GenAI to support your learning and completion of this task? In your response, address the following points: a. What GenAI tool(s) did you use, and which parts of the task did they assist with? b. How did GenAI support your understanding, confidence, efficiency, or quality of work? c. What challenges, limitations, or inaccuracies did you encounter, and how did you check or improve the GenAI output? d. How did you ensure that the final submission reflected your own understanding, decisions, and responsible use of GenAI?	a. I used Perplexity (Claude Sonnet) primarily to help break the task down into smaller, manageable parts and to clarify what each of the 14 questions was actually asking me to do. This helped me plan my approach to the notebook in a structured way before I started coding. b. The tool helped me search for relevant concepts and pandas syntax more precisely, since I could ask targeted questions about specific methods like <code>groupby()</code>, <code>pivot()</code>, and standard deviation interpretation. This improved my efficiency in locating the right approach, though the actual coding, execution, and interpretation of results were carried out independently by me. c. When I asked the AI tool to help draft parts of this report directly, it declined to generate personal reflections, insights, or analysis on my behalf, explaining that these sections needed to reflect my own understanding and decisions rather than AI-generated content. This meant I had to provide my own actual results and interpretations from my
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ITEM	RESPONSE (THE VIDEO LINK/NA)
	notebook, which the tool then only helped me structure and phrase clearly. I verified all factual claims and figures myself against my own executed notebook output before finalising any wording. d. I ensured this by running and verifying every part of my notebook independently first, cross-checking my results against the provided P2-Solution notebook, and only using AI assistance for task breakdown, syntax clarification, and language refinement rather than for generating my analysis or conclusions. All insights and interpretations described in this report are based on my own review of the actual output I produced.

10. ETHICAL ASPECTS OF AI AND DATA SCIENCE

Choose an AI or Data Science application (e.g., ChatGPT, facial recognition, recommendation systems, healthcare AI, autonomous vehicles, or social media algorithms) and discuss: - One ethical concern associated with its use. - The potential impact of this concern on individuals or society. - One strategy that could reduce or address the ethical risk.	- A significant ethical concern with recommendation systems is algorithmic bias leading to filter bubbles, where collaborative filtering algorithms disproportionately promote already-popular items while under-representing less popular or minority-relevant content, a phenomenon known as popularity bias. - This narrows the diversity of information and products users are exposed to over time, reinforcing existing preferences rather than introducing new perspectives, and can further marginalise already underrepresented groups, since data points at the margins of the distribution often correspond to minority users. At a societal level, this contributes to echo chambers, where users are consistently exposed to content aligning with their existing viewpoints, reducing exposure to diverse opinions and potentially reinforcing polarisation. - One effective mitigation strategy is diversity-aware recommendation design, where algorithms are deliberately adjusted to incorporate diversity metrics alongside accuracy, ensuring that less popular or minority-relevant items are still surfaced to users. Research has shown that diversified recommendation strategies can measurably reduce filter bubble effects compared to non-diversified approaches. This can be complemented by transparency mechanisms, such as those
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ITEM	RESPONSE (THE VIDEO LINK/NA)
	recommended by IEEE, which call for AI systems to make it possible to discover how and why a particular recommendation was generated, enabling users and auditors to identify and challenge biased outcomes.

11.ACKNOWLEDGEMENT

BY SUBMITTING THIS REPORT, I ACKNOWLEDGE THAT:

- MY RESPONSES ARE ACCURATE AND ARE MY OWN WORDS.
- I HAVE MET ALL THE BASIC REQUIREMENTS OF THE TASK (LISTED IN SECTION 6).
- I HAVE READ AND FULLY UNDERSTOOD THE ASSESSMENT GUIDELINES OF THE UNIT.
- THIS REPORT DOES NOT EXCEED 3 PAGES.
- THIS REPORT DOES NOT INCLUDE CODE EXCEPT BRIEF REFERENCES TO FUNCTION NAMES OR DIFFERENT PARTS OF THE TASK'S NOTEBOOK.
- MY SUBMISSION DOES NOT CONTAIN ANY CREDENTIALS (E.G., PASSWORD, API KEY, ETC) OR PERSONAL INFORMATION.

IMPORTANT NOTE 3: IF YOU HAVE ANSWERED NO TO ANY OF THE QUESTIONS IN SECTION 6, PLEASE RECONSIDER SUBMITTING YOUR REPORT; ASK FOR HELP FROM YOUR TUTOR.

ADD YOUR NAME AND SIGNATURE HERE: JAHAN GARG

Jahan